Greater Effect of East versus West Travel on Jet Lag, Sleep, and Team Sport Performance

PETER M. FOWLER[1], WADE KNEZ[2], STEPHEN CROWCROFT[3], AMY E. MENDHAM[4], JOANNA MILLER[5], CHARLIE SARGENT[6], SHONA HALSON[5], and ROB DUFFIELD[3]

[1]Athlete Health and Performance Research Centre, Aspetar Orthopaedic and Sports Medicine Hospital, Doha, QATAR; [2]The University of Queensland, School of Human Movement and Nutrition Sciences, QLD, AUSTRALIA; [3]Sport and Exercise Discipline Group, University of Technology Sydney, NSW, AUSTRALIA; [4]Division of Exercise Science and Sports Medicine, University of Cape Town, Cape Town, SOUTH AFRICA; [5]Physiology, Australian Institute of Sport, Canberra, ACT, AUSTRALIA; and [6]Appleton Institute, Central Queensland University, Adelaide, SA, AUSTRALIA

ABSTRACT

FOWLER, P. M., W. KNEZ, S. CROWCROFT, A. E. MENDHAM, J. MILLER, C. SARGENT, S. HALSON, and R. DUFFIELD. Greater Effect of East versus West Travel on Jet Lag, Sleep, and Team Sport Performance. *Med. Sci. Sports Exerc.*, Vol. 49, No. 12, pp. 2548–2561, 2017. **Purpose**: This study aimed to determine the recovery timeline of sleep, subjective jet lag and fatigue, and team sport physical performance after east and west long-haul travel. **Methods**: Ten physically trained men underwent testing at 0900 h and 1700 h local time on four consecutive days 2 wk before outbound travel (BASE), and the first 4 d after 21 h of outbound (WEST) and return (EAST) air travel across eight time zones between Australia and Qatar. Data collection included performance (countermovement jump, 20-m sprint, and Yo-Yo intermittent recovery level 1 [YYIR1] test) and perceptual (jet lag, motivation, perceived exertion, and physical feeling) measures. In addition, sleep was measured via wrist activity monitors and self-report diaries throughout the aforementioned data collection periods. **Results**: Compared with the corresponding day at BASE, the reduction in YYIR1 distance after EAST was significantly different from the increase in WEST on day 1 after travel ($P < 0.001$). On day 2, significantly slower 20-m sprint times were detected in EAST compared with WEST ($P = 0.03$), with large effect sizes (ES) also indicating a greater reduction in YYIR1 distance in EAST compared with WEST ($d = 1.06$). Mean sleep onset and offset were significantly later and mean time in bed and sleep duration were significantly reduced across the 4 d in EAST compared with BASE and WEST ($P < 0.05$). Lastly, mean jet lag, fatigue, and motivation ratings across the 4 d were significantly worse in EAST compared with BASE and WEST ($P < 0.05$) and WEST compared with BASE ($P < 0.05$). **Conclusions**: Long-haul transmeridian travel can impede team sport physical performance. Specifically, east travel has a greater detrimental effect on sleep, subjective jet lag, fatigue, and motivation. Consequently, maximal and intermittent sprint performance is also reduced after east travel, particularly within 72 h after arrival. **Key Words:** SOCCER, TEAM SPORT, INTERMITTENT EXERCISE, LONG-HAUL TRAVEL, TRAVEL FATIGUE, SLEEP DISRUPTION

Professional team sport athletes regularly undertake long-haul transmeridian travel for competition, training camps, and/or preseason tours. Together with financial considerations, these demanding schedules dictate travel itineraries and consequently the proximity of arrival before the commencement of competition. Although optimal readiness to perform is required in the days after arrival, limited evidence exists describing the effect of an episode of long-haul travel on team sport–specific physical performance (1,22). Consequently, inferences from an understanding of chronobiology obtained through stringently controlled laboratory studies are used to inform field-based travel strategies. For example, on the basis of the premise that it is easier to delay than advance circadian rhythms, in theory, jet lag symptoms should be worse when traveling east compared with west (34). Although this has implications for athlete preparedness after travel, there is currently no information on the posttravel timeline of recovery of physical performance for team sports or a comparison of this timeline between east and west travel.

The interrelated effects of both jet lag and travel fatigue are present after long-haul transmeridian travel, with increased daytime fatigue, reduced motivation, and sleep disruption commonly observed (2,14,37). Upon arrival, endogenous circadian rhythms in body temperature and melatonin are misaligned with external cues at the destination (8,23,25). Until these rhythms align with the new time zone, sleep, fatigue, motivation, and ensuing physical performance may be negatively affected (22,25). Moreover, a change in the rhythm of a morning nadir and late afternoon peak in physical performance may occur, with both physical and mental performance capacities reduced outside of this circadian peak "window" (22). In addition, conditions encountered during travel, such as the uncomfortable seating arrangements, noise

Address for correspondence: Peter Fowler, Ph.D., Athlete Health and Performance Research Centre, Aspetar Orthopaedic and Sports Medicine Hospital, Sports City Street, Doha, Qatar; E-mail: peter.fowler@aspetar.com.
Submitted for publication May 2017.
Accepted for publication July 2017.

0195-9131/17/4912-2548/0
MEDICINE & SCIENCE IN SPORTS & EXERCISE®
Copyright © 2017 by the American College of Sports Medicine
DOI: 10.1249/MSS.0000000000001374

APPLIED SCIENCES

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

levels, and timing of stopovers, may also disrupt sleep and exacerbate mood states (6,11,14). Although these demands may contribute to performance reductions in combination with circadian misalignment, data on the recovery timeline of and the time of day influence on sports-specific performance after an acute episode of long-haul travel remain limited.

Changes in the grip strength have consistently been reported after a long-haul transmeridian travel in both athletic (23,25) and nonathletic (8) populations. Specifically, a change in the performance rhythm of a morning nadir and a late afternoon peak (local time) occurs and grip strength is reduced outside of its circadian peak window (8,23,25). Reduced countermovement jump (CMJ) performance has also been observed after 24-h travel east across eight time zones in national team skeleton athletes (5). However, although the use of grip strength and jump tests as performance measures may assist with the logistics of testing athletes around travel and competition, they have limited ecological relevance to physical performance in team sports (22). As a consequence, most current travel advice is based on stringently controlled laboratory-based research, which to date has had poor translation to high-performance athletes in the field (1,33). For example, the rate of adaptation to a new time zone is estimated as half a day per hour of the time difference west and one day per hour of the time difference east (9,26). Although these rates of adaptation have implications for team sport physical performance, the timeline of performance recovery after travel in either direction is yet to be elucidated (9,22,26).

As a result of this limited understanding of the performance recovery timeline after long-haul transmeridian travel, many practitioners currently rely on personal experience or anecdotal evidence for travel schedules and training prescription after travel. Thus, the aim of the present study was to compare the timeline of recovery of sleep, subjective jet lag, fatigue, and physical performance measures related to team sports after east and west long-haul transmeridian travel. It was hypothesized that physical performance, sleep, jet lag, and fatigue would be worse after travel, and that east travel would cause greater disruption to these responses compared with west travel.

METHODS

Participants

Thirty students from a local university (Sydney, Australia) were screened through a general health questionnaire and an interview with a member of the research team for inclusion in the study. Eleven were excluded on the basis of the following criteria: female, current smoker, sleep disorder, sleep medication use or illness, extreme morning or evening diurnal preference, insufficient fitness level (unable to obtain level 16 on the Yo-Yo intermittent recovery level 1 [YYIR1] test), and shiftwork or transmeridian travel across more than two time zones undertaken in the month before data collection. Of the 19 available participants, 10 healthy, physically trained men were recruited to participate (mean ± SD: age, 20.6 ± 2.7 yr; height, 179.1 ± 5.4 cm; and body mass, 77.05 ± 7.83 kg). All participants had recent training history in a range of athletic events, including running, football, rugby, and soccer, and were involved in some form of physical (aerobic and strength) training at least 2–3 times per week. Before the commencement of the study, participants were informed of any associated risks and provided verbal and written informed consent. The study was approved by the Anti-Doping Lab Qatar Institutional Ethics Review Board (IRB No. EXT2014000003).

Experimental Design

Participants completed a minimum of two familiarization sessions with all experimental measures and procedures. Performance data were collected at 0900 h and 1700 h local time (deemed ecologically relevant to training and competition) on four consecutive days in the 2 wk before outbound travel as a baseline (BASE), and the first 4 d after long-haul transmeridian travel from Australia to Qatar (WEST; Fig. 1). After a 4-d wash-out, data were collected at the same time of day, 2 d before (PRE-EAST) and on the first 4 d after return travel from Qatar to Australia (EAST). In addition, sleep was measured throughout the aforementioned data collection periods and perceptual data were collected immediately before all performance testing sessions. Participants abstained from caffeine, alcohol, and additional strenuous activity for 24 h before and during each data collection period. Considering the outside daytime temperatures in Qatar (~30°C–35°C) at the time of data collection (September/October) and the potential influence of heat acclimation on performance, participants' heat exposure was minimized to less than 1-h incidental passive heat exposure per day. Food and fluid intake was documented throughout in a food diary, with participants instructed to replicate their intake during BASE as closely as possible during WEST and EAST. Participants were provided with a standardized 1.5–2.0 $g \cdot kg^{-1}$ body mass of carbohydrate, including 600 mL of fluid (Gatorade™; PepsiCo, Chicago, IL) immediately after all performance testing sessions. Before all performance testing, urine specific gravity (USG) was assessed (Digital Refractometer, Atago, WA) to determine hydration status from a midstream urine sample.

Travel

The departure and arrival times for WEST (outbound travel) were 1930 h Australian Eastern Standard Time (AEST) and 0855 h the next day Arabia Standard Time (AST [AEST −8 h]). In total, there were three flights, with 21 h of travel (17 h of total flight duration, 4 h of total transit time) west across eight time zones:

1. Sydney, Australia (1930 h local time) to Melbourne, Australia (2105 h)
2. Melbourne, Australia (2240 h) to Abu Dhabi, United Arab Emirates (0645 h + 1 d)
3. Abu Dhabi, United Arab Emirates (0855 h) to Doha, Qatar (0855 h)

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

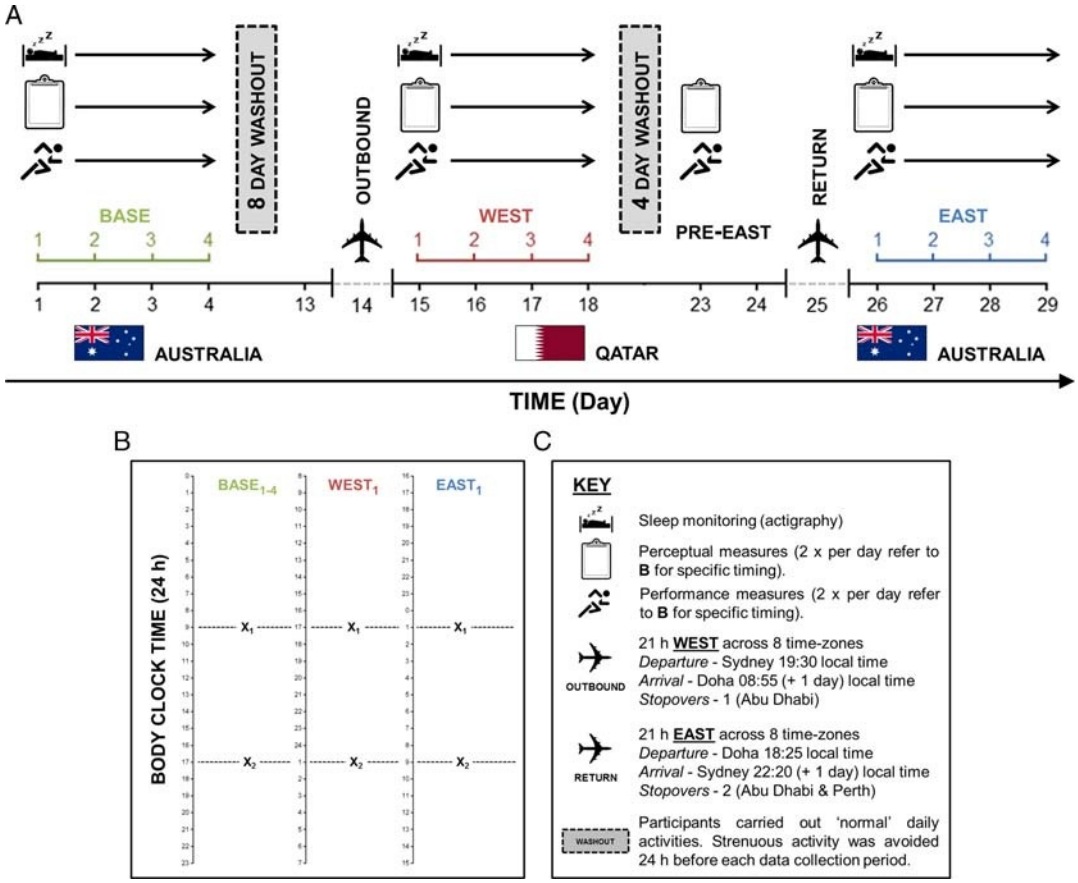

FIGURE 1—Schematic outline of the study design including the timeline of data collection (A), estimated body clock time data collected on the first day after WEST and EAST (B), and specific details of the measures, travel, and washout (C).

The departure and arrival times for EAST (return travel) were 1825 h AST and 2220 h the next day AEST (AST +8 h). In total, there were three flights, with 21 h of travel (17 h of total flight duration, 4 h of total transit time) east across eight time zones:

1. Doha, Qatar (1825 h local time) to Abu Dhabi, United Arab Emirates (2020 h)
2. Abu Dhabi, United Arab Emirates (2220 h) to Perth, Australia (1350 h + 1 d)
3. Perth, Australia (1600 h) to Sydney, Australia (2220 h)

Participants traveled in economy class for all flights and were given no instructions by the research team regarding behaviors during travel (e.g., when to eat and sleep, how much fluid to consume, and how often they should walk around the cabin). To match as closely as possible the duration (h) between arrival and initial testing in EAST and WEST, because of the differences in local arrival times, none of the measures outlined below were collected on day 1 at 0900 h in EAST.

Experimental Procedures

Physical performance. All performance testing sessions in both locations were performed indoors on a basketball court in a temperate, air-conditioned environment (~20°C–25°C), with the same equipment and research team. A warm-up consisting of 5 min of standardized submaximal running followed by 10 min of general whole-body movements was completed at the beginning of all performance testing sessions (32). Participants performed a CMJ test (32) at both 0900 and 1700 h, along with a 20-m sprint test and the YYIR1 test (20) at 1700 h only. These tests are commonly used to assess the physical performance levels of team sport athletes, and participants were blinded from their results in all performance tests until the end of the study.

Jump height was measured using a linear position transducer (GymAware; Kinetic Performance Technologies, Canberra, ACT, Australia) sampling at 50 Hz, which was placed between the participant's feet on the floor and attached to a belt secured around the waist. The linear transducer transmitted displacement-time data to a hand-held unit, which was subsequently analyzed using commercially available software (GymAware; Kinetic Performance Technologies). Participants performed two sets of three maximal CMJ values using a self-selected depth, with a minimum of 3-min recovery between each set. Peak power and force and jump height were determined for each jump, and after the removal of outliers (>3 SD values away from the mean), an average of the six jumps was calculated and used for analyses.

Three maximal 20-m sprints were performed with a minimum of 3-min recovery between each. Splits were measured

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

at 5 and 20 m using a single-beam infrared timing gate system (Microgate, Bolanzo, Italy). The fastest 5- and 20-m sprint times were included in ensuing analyses. YYIR1 performance was determined by total distance covered at the point of volitional exhaustion, which has been identified as a valid and reliable measure of team sport physical performance capacity (20). A modified version of the YYIR1 was used with feedback regarding the level removed from the test audio mp3. Given the removal of audio feedback, a stopwatch was used to record the time elapsed at volitional exhaustion for each participant, which was later converted to total distance covered.

Sleep. Sleep patterns were monitored using self-report diaries and wrist activity monitors (Actiwatch-64; Philips Respironics, Bend, OR). According to previously described methods, data from the sleep diaries and activity monitors were used to determine when participants were awake and asleep (29). All time records were scored as wake unless (i) the sleep diary indicated that the participant was lying down attempting to sleep and (ii) the activity counts from the monitor were sufficiently low to indicate that the participant was immobile (i.e., where the weighted activity count for an epoch fell below the defined threshold). In this study, sensitivity was set at medium, which corresponds to a threshold activity count of 40 (35). When these two conditions were satisfied simultaneously, time was scored as sleep. This scoring process was conducted using a Philips Respironics' Actiwatch algorithm, which has been used to quantify sleep/wake behavior in airline pilots and long-haul truck drivers (7,20). The following variables were derived from the sleep diary and activity monitor data;

- Time in bed (h:min): the period between going to bed and getting up
- Sleep onset (hh:mm): the time at which a participant first fell asleep after going to bed
- Sleep offset (hh:mm): the time at which a participant last woke before getting up
- Sleep duration (h:min): the amount of time spent in bed asleep
- Sleep efficiency (%): sleep duration expressed as a percentage of time in bed
- Daytime nap duration (min): the amount of time spent in bed asleep during a daytime nap
- Cumulative sleep duration (h:min): the sum of the sleep obtained at night and any sleep obtained the following day during a daytime nap(s)

NB: To isolate the effect of long-haul travel on nighttime sleep, participants were instructed to avoid daytime napping throughout the study. Analysis of the sleep diaries and activity monitors indicated, however, that some participants were unable to avoid daytime napping.

Perceptual. Unless stated otherwise, all perceptual measures were entered into a Microsoft Excel spreadsheet on a laptop computer immediately before the physical performance testing. Each participant sat at a separate desk, had no access to previous responses, used an individual laptop, and wore noise canceling headphones to minimize external distractions and peer influence on their responses (4).

The Liverpool John Moore's University jet lag questionnaire was used to assess participants' subjective ratings of jet lag (37). The specific times of assessment for each subscale were as follows: 1) jet lag (one question), 0900 and 1700 h; 2) sleep (five questions), 0900 h only; 3) fatigue (one question), 0900 and 1700 h; 4) diet (three questions), 1700 h only; 5) mental performance and mood (three questions), 1700 h only; and 6) bowel movement (two questions), 1700 h only. Following a method previously outlined (33), data with a "negative outcome" were given a positive score and data with a "positive outcome" were given a negative score, before being pooled for summation into five categories: (a) jet lag, (b) sleep, (c) function (c.1. fatigue + c.2. mental performance and mood), (d) diet, and (e) bowel movement. A greater overall value indicates worse symptoms (i.e., a positive number indicates a worse than "normal" response and a negative number indicates a better than "normal" response). At both 0900 and 1700 h, participants' general level of motivation was assessed on a Likert scale from 0 ("none") to 4 ("very motivated") in 0.5-point increments. Lastly, approximately 30 min after the completion of the physical performance tests (1700 h only), a session rating of perceived exertion (sRPE) (10) and physical feeling (18) were obtained from participants.

Statistical Analysis

Unless specified, data are presented as mean ± SD. Normality of the observed data was assessed using quantile–quantile (Q–Q) plots and was deemed plausible in all instances, with data presented as mean ± SD. Initially, comparisons between BASE day 1 and PRE-EAST day 1 were made for all variables using linear-mixed models and standardized ES (Cohen's *d*) analysis. Because no significant ($P < 0.05$) differences were observed and all ES values (Cohen's *d*) were small ($d < 0.40$), raw change in all performance variables on each of the 4 d after WEST and EAST were calculated from the corresponding day at BASE (i.e., day 1 EAST was compared with day 1 BASE). As such, PRE-EAST measures were used as an internal control to ensure no training or fatigue effect had occurred before return travel. Raw values on each of the 4 d at BASE, WEST, and EAST were used for all sleep and perceptual variables. Differences between condition (travel direction) and interactions between condition and time (day and time of day) for the raw change in performance variables and raw values for sleep and perceptual variables were analyzed using linear-mixed models. This type of analysis is preferred because it (i) allows for missing data, (ii) can accurately model different covariate structures for repeated-measures data, and (iii) can model between-subject variability (36). The most appropriate model for each variable was chosen using the smallest Akaike information criterion in accordance with the principal of parsimony. Where significance was obtained ($P < 0.05$), least significant difference *post hoc* tests were used. All statistical analyses were performed using

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

SPSS (Version 21; IBM, Armonk, NY). Furthermore, standardized ES analysis was used to interpret the magnitude of differences between conditions and over time. Because of the substantial amount of statistical analyses performed, only large ES values ($d > 0.90$) are reported.

RESULTS

To aid clarity, results are structured within each ensuing paragraph as follows: (a) differences between travel direction (BASE, WEST, and EAST), (b) interactions between travel direction and day (1–4), and (c) interactions between time of day (0900 and 1700 h) with travel direction and day. Of note, 0900 and 1700 h *local time* would have corresponded to approximately 1700 and 0100 h *body clock time* on day 1 after WEST and 0100 and 0900 h *body clock time* after EAST (Fig. 1).

Physical Performance

Results for CMJ performance variables are presented in Table 1. A significant main effect of travel direction was observed for peak force ($F_{1,84} = 3.94$, $P = 0.04$), with a greater mean reduction across the 4 d in EAST compared with WEST ($P = 0.03$). Furthermore, to be outlined in the following section, significant interactions between travel direction, day, and time of day were evident for both peak power ($F_{5,86} = 4.59$, $P = 0.001$) and peak force ($F_{5,83} = 4.70$, $P = 0.001$).

AM testing (0900 h local time, ~1700 h WEST, ~0100 h EAST)

a. Travel direction

A greater reduction in peak power was detected in WEST compared with EAST on day 2 ($P > 0.05$, $d > 0.90$).

b. Day

Compared with day 1, a significantly greater reduction in peak force was observed on days 2 and 4 in WEST ($P < 0.05$).

PM testing (1700 h local time, ~0100 h WEST, ~0900 h EAST)

a. Travel direction

Although no significant differences were detected between travel directions, large ES indicated that the increase in peak power and peak force on day 1 in WEST was different from the reduction observed in EAST ($P > 0.05$, $d > 0.90$).

b. Day

The increase in peak power on day 4 was significantly different from the reduction on days 2 and 3 in WEST and EAST ($P < 0.05$). The increase in peak force on day 1 was significantly different from the reduction on days 2 and 3 in WEST ($P < 0.05$), and the reduction in peak force on day 3 was significantly different from all other days in WEST and days 1 and 4 in EAST ($P < 0.05$).

c. Time of day

On day 3, the reduction in peak force at 1700 h was significantly greater than the reduction at 0900 h in WEST ($P < 0.05$). On day 4, the increase in peak power at 1700 h was significantly different from the reduction at 0900 h in both WEST and EAST, and the increase in peak force at 1700 h was significantly different from the reduction at 0900 h in WEST ($P < 0.05$).

Twenty-meter sprint and YYIR1 results are presented in Figure 2. A significant interaction between travel direction and day was observed for 20-m sprint ($F_{6,49} = 3.46$, $P = 0.006$) and YYIR1 ($F_{6,39} = 5.96$, $P < 0.001$) performance, respectively.

PM testing (1700 h local time, ~0100 h WEST, ~0900 h EAST)

a. Travel direction

A significantly slower 20-m sprint time was detected on day 2 in EAST compared with WEST ($P = 0.03$). On day 1, the reduction in YYIR1 distance in EAST was significantly different from the increase in WEST ($P < 0.001$). Large ES indicated a greater reduction in YYIR1 distance in EAST compared with WEST on day 2 ($d = 1.06$). Conversely, the increase in YYIR1 distance on day 4 in EAST was greater than that in WEST ($d = 1.00$).

b. Day

Five-meter sprint time was significantly slower on days 1 and 2 compared with day 4 in EAST ($P < 0.05$), with large ES also suggesting that it was slower on day 3 ($d = 1.25$). Similarly, 20-m sprint time was significantly slower on days 1 and 2 compared with days 3 and 4 in EAST ($P < 0.05$). The reduction in YYIR1 distance on days 1 and 2 in EAST was significantly different from the increase on days 3 and 4 ($P < 0.05$), whereas the increase in YYIR1 distance on day 1 was significantly greater than that on days 2 and 4 in WEST ($P < 0.05$).

Sleep

Results for all sleep variables are presented in Table 2. A significant main effect of travel direction was observed for time in bed ($F_{2,71} = 15.57$, $P < 0.001$), sleep onset ($F_{2,70} = 36.69$, $P < 0.001$), sleep offset ($F_{2,67} = 4.75$, $P = 0.012$), and sleep duration ($F_{2,70} = 13.03$, $P < 0.001$). A significant interaction between travel direction and day was detected for sleep onset ($F_{11,49} = 4.00$, $P < 0.001$), sleep offset ($F_{11,41} = 5.78$, $P < 0.001$), and sleep duration ($F_{11,48} = 2.11$, $P = 0.037$).

a. Travel direction

Mean sleep onset and mean offset were significantly later and mean time in bed and mean sleep duration were

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

TABLE 1. Change from BASE in CMJ performance variables in WEST and EAST.

		Day 1		Day 2		Day 3		Day 4		Mean		
		Time										
		0900 h	1700 h	0900 h	1700 h	0900 h	1700 h	0900 h	1700 h	0900 h	1700 h	Overall
ΔPeak power, W[c]												
TE 295	WEST	−98 ± 457 1392 (−523 to 869)	60 ± 400 1416 (−664 to 752)	−447 ± 354[j,e] 1197 (−882 to 315)	−227 ± 651 2162 (−1759 to 403)	−80 ± 431 1494 (−803 to 691)	−308 ± 337 1062 (−880 to 182)	−454 ± 310[n] 904 (−971 to −67)	222 ± 479[e,g,p] 1350 (−244 to 1106)	−270 ± 136 1840 (−971 to 869)	−109 ± 369 2865 (−1759 to 1106)	−177 ± 469 2865 (−1759 to 1106)
	EAST		−198 ± 377[j] 1386 (−878 to 508)	−120 ± 287 965 (−543 to 422)	−394 ± 637 1955 (−1653 to 302)	−135 ± 358 1156 (−592 to 564)	−443 ± 584[k] 1763 (−1399 to 364)	−331 ± 364 1250 (−815 to 435)	177 ± 637[e,g,l] 1960 (−698 to 1262)	−196 ± 185 1379 (−815 to 564)	−223 ± 367 2915 (−1653 to 1262)	−212 ± 495 2915 (−1653 to 1262)
ΔPeak force, N[a,b,c]												
TE 100	WEST	−13 ± 158[m] 456 (−267 to 189)	29 ± 124[g,o] 396 (−149 to 247)	−182 ± 147[i,l] 477 (−469 to 8)	−109 ± 185 527 (−366 to 161)	−100 ± 92 314 (−244 to 70)	−244 ± 137[e,f] 438 (−443 to −5)	−187 ± 141[i,m] 446 (−410 to 36)	40 ± 205[e,f] 551 (−191 to 360)	−121 ± 79 658 (−469 to 189)	−85 ± 105 3803 (−443 to 360)	−99 ± 173 829 (−469 to 360)
	EAST		−72 ± 148[j] 442 (−341 to 101)	−156 ± 148 494 (−413 to 81)	−143 ± 168 465 (−393 to 72)	−148 ± 126 329 (−343 to −14)	−267 ± 87[h,l] 307 (−367 to −60)	−179 ± 136 394 (−375 to 19)	−58 ± 187 621 (−361 to 260)	−162 ± 67 494 (−413 to 81)	−135 ± 96 653 (−393 to 260)	−148 ± 152[d] 673 (−413 to 260)
ΔHeight, cm												
TE 1.1	WEST	−0.9 ± 2.8 8.9 (−3.5 to 5.4)	−1.0 ± 2.2 6.9 (−3.6 to 3.3)	−2.1 ± 2.2 7.0 (−4.2 to 2.8)	−2.4 ± 4.2 14.3 (−10.0 to 4.3)	−1.9 ± 2.8 8.0 (−6.3 to 1.7)	−3.0 ± 1.9 5.9 (−6.5 to −0.6)	−1.2 ± 2.6 6.9 (−5.1 to 1.8)	−1.9 ± 2.5 5.9 (−5.1 to 0.8)	−1.5 ± 1.5 11.7 (−6.3 to 5.4)	−2.2 ± 2.2 14.3 (−10.0 to 4.3)	−1.8 ± 2.6 15.4 (−10.0 to 5.4)
	EAST		−1.6 ± 2.9 9.6 (−6.1 to 3.5)	−1.6 ± 3.1 9.1 (−5.0 to 4.1)	−1.6 ± 3.3 12.0 (−7.7 to 4.3)	−1.9 ± 3.5 11.7 (−6.1 to 5.6)	−1.8 ± 2.5 8.1 (−5.2 to 2.9)	−0.7 ± 2.0 5.2 (−2.8 to 2.4)	−0.9 ± 2.6 7.1 (−4.0 to 3.1)	−1.4 ± 2.5 11.7 (−6.1 to 5.6)	−1.5 ± 2.4 12.0 (−7.7 to 4.3)	−1.4 ± 2.7 13.3 (−7.7 to 5.6)

Data are presented as mean ± SD and range (minimum–maximum).
[a]Significant main effect for travel direction ($P < 0.05$).
[b]Significant interaction between travel direction and day ($P < 0.05$).
[c]Significant interaction between travel direction, day, and time of day ($P < 0.05$).
[d]EAST significantly different from WEST ($P < 0.05$).
[e]Significantly different from 0900 h ($P < 0.05$).
[f]Significantly different from all other days ($P < 0.05$).
[g]Significantly different from days 2 and 3 ($P < 0.05$).
[h]Significantly different from days 1 and 4 ($P < 0.05$).
[i]Significantly different from day 1 ($P < 0.05$).
[j]Large ES indicating that EAST is different from WEST ($d > 0.90$).
[k]Large ES indicating difference from 0900 h ($d > 0.90$).
[l]Large ES indicating difference from all other days ($d > 0.90$).
[m]Large ES indicating difference from days 1 and 2 ($d > 0.90$).
[n]Large ES indicating difference from days 1 and 3 ($d > 0.90$).
[o]Large ES indicating difference from days 2 and 3 ($d > 0.90$).
TE, typical error.

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

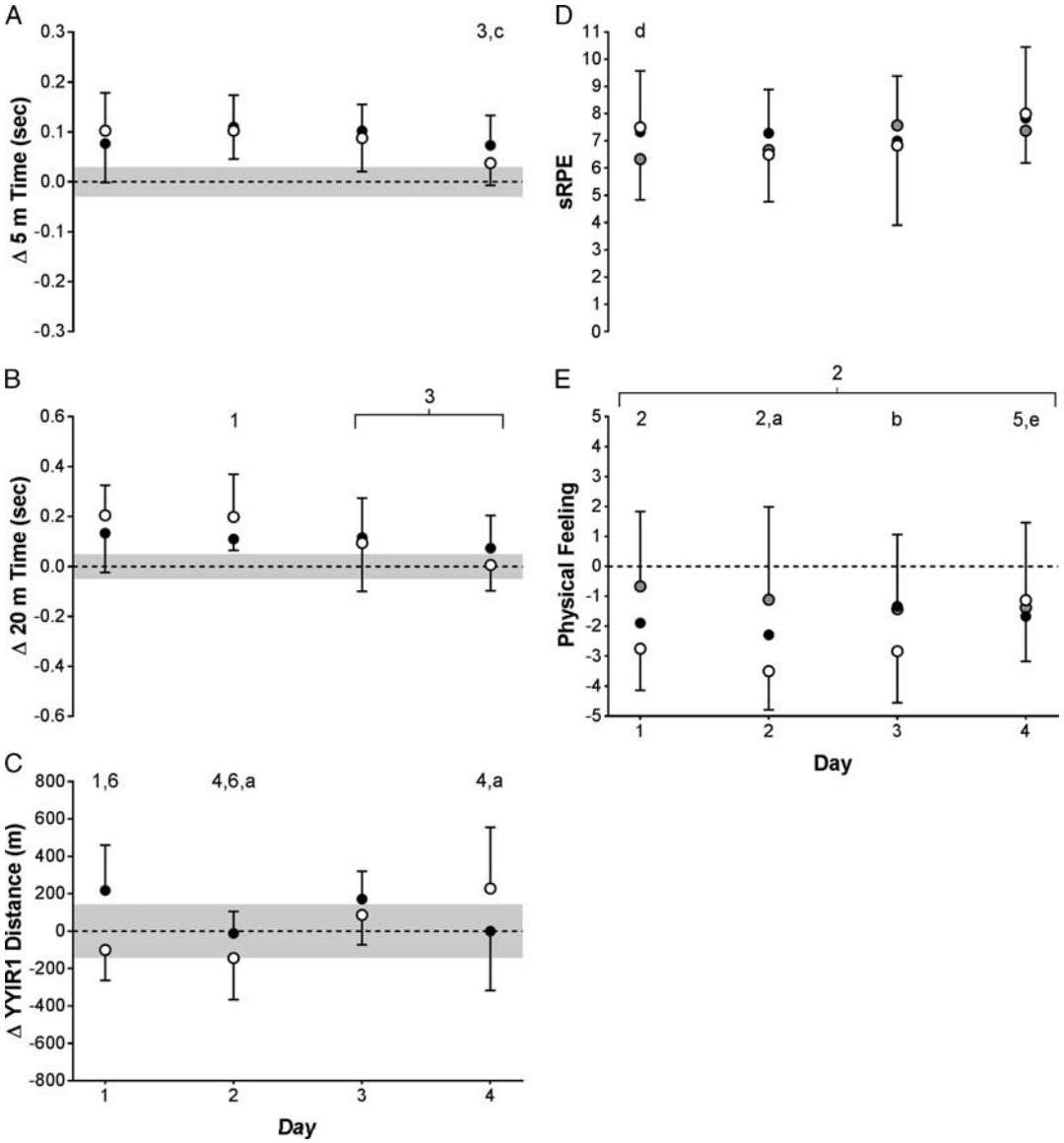

FIGURE 2—Mean ± SD changes from BASE in sprint (A and B) and YYIR1 (C) performance in WEST (black circles) and EAST (white circles). Gray shaded area indicates the typical error of the measure. Mean ± SD responses for sRPE (D) and physical feeling (E) during BASE (gray circles), WEST and EAST. [1]EAST significantly different from WEST ($P < 0.05$). [2]EAST significantly different from BASE ($P < 0.05$). [3]Significantly different from days 1 and 2 in EAST ($P < 0.05$). [4]Significantly different from day 1 in WEST ($P < 0.05$). [5]Significantly different from day 1 in BASE and EAST ($P < 0.05$). [6]Significantly different from days 3 and 4 in EAST ($P < 0.05$). [a]Large ES indicating that EAST is different from WEST ($d > 0.90$). [b]Large ES indicating that EAST is different from BASE and WEST ($d > 0.90$). [c]Large ES indicating difference on day 3 in EAST ($d > 0.90$). [d]Large ES indicating difference from days 2 and 3 in BASE ($d > 0.90$). [e]Large ES indicating difference from all other days in EAST ($d > 0.90$).

significantly reduced across the 4 d in EAST compared with BASE and WEST ($P < 0.05$). Sleep onset and sleep offset were significantly later and time in bed and sleep duration were significantly reduced on the day of arrival in EAST compared with WEST ($P < 0.05$). Similarly, sleep onset was significantly later and time in bed and sleep duration were significantly reduced on days 1 and 2 in EAST compared with BASE and WEST ($P < 0.05$). Sleep onset was also significantly later on day 4 in EAST compared with BASE and WEST ($P < 0.05$). Large ES suggested that sleep offset was later on days 2 and 3 in WEST compared with BASE, and sleep efficiency was reduced on day 1 in WEST and EAST compared with BASE ($d > 0.90$).

b. Day

In WEST, sleep onset was significantly earlier and time in bed and sleep duration were significantly greater on the day of arrival compared with all other days ($P < 0.05$). Similarly, time in bed and sleep duration were significantly greater on the day of arrival compared with days 1 and 2 in EAST ($P < 0.05$). Moreover, sleep onset was significantly earlier on day 3 compared with day of arrival and day 4, sleep offset was significantly later on

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

TABLE 2. Sleep patterns, quantity, and quality during BASE, WEST, and EAST.

	Arrival	Day 1	Day 2	Day 3	Day 4	Mean
Time in bed, h:min[a]						
BASE		07:14 ± 01:01 03:16 (05:21–08:37)	07:33 ± 01:23 04:34 (04:46–09:20)	07:22 ± 00:55 03:15 (05:52–09:07)	07:48 ± 01:00 03:09 (06:05–09:14)	07:31 ± 00:49 03:01 (05:59–09:01)
WEST	08:33 ± 00:50[e] 02:30 (07:11–09:41)	07:21 ± 00:32 01:39 (06:40–08:19)	07:22 ± 00:42 02:01 (06:20–08:21)	07:11 ± 00:23 01:12 (06:37–07:49)	07:48 ± 00:20[d] 00:57 (07:16–08:13)	07:37 ± 00:21 01:11 (06:58–08:09)
EAST	07:20 ± 00:40[c] 02:06 (06:14–08:20)	06:13 ± 01:27[d,f] 04:31 (03:06–07:37)	06:11 ± 01:43[d,f] 04:49 (03:33–08:22)	06:38 ± 01:27 04:29 (04:00–08:29)	07:06 ± 01:20[g] 03:21 (05:47–09:08)	06:36 ± 00:53[d] 02:30 (05:13–07:44)
Sleep onset, hh:mm[a,b]						
BASE		23:09 ± 01:11 03:16 (21:42–00:58)	22:57 ± 01:27 04:53 (21:00–01:53)	23:00 ± 01:06 04:12 (21:00–01:12)	23:57 ± 01:19[e] 04:04 (22:02–02:06)	23:11 ± 01:15 04:17 (21:14–01:32)
WEST	22:00 ± 00:52[e] 02:31 (21:05–23:36)	23:12 ± 00:43 01:46 (22:27–00:13)	23:23 ± 00:41 02:01 (22:29–00:30)	23:27 ± 00:35 01:44 (22:41–00:25)	23:02 ± 00:19 00:58 (22:27–23:25)	23:02 ± 00:30 01:32 (22:23–23:56)
EAST	00:57 ± 01:13[c] 03:01 (23:51–02:52)	00:11 ± 01:23[d] 03:51 (22:55–02:46)	00:24 ± 01:38[d] 04:17 (22:52–03:09)	23:40 ± 01:43[f] 05:10 (21:50–03:00)	00:36 ± 01:25[d] 03:55 (23:08–03:03)	00:28 ± 01:11[d] 03:28 (23:10–02:38)
Sleep offset, hh:mm[a,b]						
BASE		06:16 ± 00:41 02:19 (04:51–07:10)	06:18 ± 00:27 01:37 (05:43–07:20)	06:08 ± 00:36 02:03 (05:13–07:16)	07:34 ± 01:37[e] 04:20 (05:45–10:05)	06:32 ± 00:31 01:23 (05:55–07:18)
WEST	06:34 ± 00:21 01:14 (05:44–06:58)	06:32 ± 00:27 01:12 (05:47–06:59)	06:44 ± 00:15[c] 00:46 (06:22–07:08)	06:37 ± 00:30[c] 01:34 (05:56–07:30)	06:51 ± 00:21 01:00 (06:31–07:31)	06:39 ± 00:15 00:50 (06:15–07:06)
EAST	08:14 ± 01:20[c,e] 04:10 (06:59–11:09)	06:24 ± 00:28 01:28 (05:51–07:19)	06:33 ± 00:29 01:18 (05:59–07:17)	06:15 ± 00:42 02:15 (05:19–07:34)	07:39 ± 01:38[e] 04:13 (05:27–09:40)	07:02 ± 00:40[d] 02:06 (06:13–08:19)
Sleep duration, h:min[a,b]						
BASE		06:25 ± 01:06 03:24 (04:35–07:59)	06:18 ± 01:16 04:33 (03:53–08:26)	06:20 ± 00:50 03:16 (04:53–08:09)	06:38 ± 00:45 02:36 (05:11–07:47)	06:28 ± 00:48 03:08 (05:03–08:11)
WEST	07:36 ± 01:00[e] 02:43 (06:23–09:06)	06:24 ± 00:36 01:58 (05:25–07:23)	06:22 ± 00:47 02:28 (05:24–07:52)	06:15 ± 00:29 01:36 (05:38–07:14)	06:53 ± 00:55 02:20 (05:39–07:59)	06:40 ± 00:36 02:00 (05:44–07:45)
EAST	06:27 ± 00:49[c] 02:39 (04:52–07:31)	05:24 ± 01:18[d,f] 03:56 (02:38–06:34)	05:18 ± 01:38[d,f] 04:48 (03:07–07:55)	05:50 ± 01:27 04:22 (03:16–07:38)	06:13 ± 01:28[g] 03:38 (04:50–08:28)	05:46 ± 00:57[d] 02:45 (04:27–07:13)
Sleep efficiency, %						
BASE		90.3 ± 3.1 9.5 (84.4–93.9)	85.7 ± 3.5 11.9 (79.5–91.5)	88.9 ± 3.7 12.5 (83.4–96.0)	87.4 ± 4.2 13.0 (79.9–92.9)	88.2 ± 3.1 26.6 (83.4–92.7)
WEST	88.7 ± 4.0 11.9 (82.0–93.9)	87.2 ± 5.1[b] 16.9 (80.2–97.1)	86.4 ± 4.3 13.6 (80.5–94.2)	87.2 ± 3.7 9.9 (83.1–93.0)	87.8 ± 8.4 19.9 (77.4–97.3)	87.0 ± 4.7 26.5 (80.6–95.4)
EAST	88.4 ± 5.4 17.6 (78.2–95.9)	86.9 ± 3.3[b] 9.8 (81.8–91.5)	86.8 ± 9.9 31.1 (63.7–94.8)	88.5 ± 3.3 8.9 (84.9–93.8)	87.7 ± 4.2 12.0 (82.8–94.8)	87.4 ± 4.5 26.4 (78.3–93.2)

Data are presented as mean ± SD and range (minimum–maximum).
[a]Significant main effect for travel direction ($P < 0.05$).
[b]Significant interaction between travel direction and day ($P < 0.05$).
[c]EAST significantly different from WEST ($P < 0.05$).
[d]EAST significantly different from BASE and WEST ($P < 0.05$).
[e]Significantly different from all other days ($P < 0.05$).
[f]Significantly different from day of arrival ($P < 0.05$).
[g]Significantly different from days 1 and 2 ($P < 0.05$).
[h]Large ES indicating that EAST is different from WEST ($d > 0.90$).
[i]Large ES indicating that EAST and WEST are different from BASE ($d > 0.90$).
[j]Large ES indicating that WEST is different from BASE.
[k]Large ES indicating difference from all other time points ($d > 0.90$).

day of arrival compared with all other days, and time in bed and sleep duration were significantly greater on day 4 compared with days 1 and 2 in EAST ($P < 0.05$). Sleep onset and offset were significantly later on day 4 compared with all other days in BASE ($P < 0.05$).

Across the 4 d, a total of two naps were recorded in BASE and EAST, and eight naps in WEST. Mean nap duration across the 4 d was 35 ± 11, 59 ± 11, and 73 ± 4 min for BASE, WEST, and EAST, respectively. However, the interaction between travel direction and day for cumulative sleep duration was not different from the aforementioned results for sleep duration.

Perceptual

Jet lag. Results for jet lag and function ratings are presented in Figure 3. A significant main effect of travel direction was observed for jet lag ($F_{2,147} = 45.29$, $P < 0.001$), function ($F_{2,185} = 15.18$, $P < 0.001$), and diet ($F_{2,190} = 7.06$, $P = 0.001$). Specifically, mean jet lag and function ratings across the 4 d were significantly worse in EAST compared with BASE and WEST, and WEST compared with BASE ($P < 0.05$). Conversely, mean diet ratings were significantly worse in WEST (3.3 ± 2.5 AU) compared with BASE (2.2 ± 2.2 AU) and EAST (1.9 ± 2.1 AU; $P < 0.01$).

AM testing (0900 h local time, ~1700 h WEST, ~0100 h EAST)

a. Travel direction

Jet lag was significantly worse in WEST compared with BASE on days 1 and 2 ($P < 0.05$), EAST compared with BASE on days 2–4 ($P < 0.05$), and EAST compared with WEST on day 3 ($P < 0.05$). Large ES indicated that jet lag was still elevated on day 4 in EAST and WEST compared with BASE ($d > 0.90$). Function ratings were significantly worse on day 2 in EAST compared with BASE and WEST ($P < 0.05$). Large ES indicated that

APPLIED SCIENCES

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

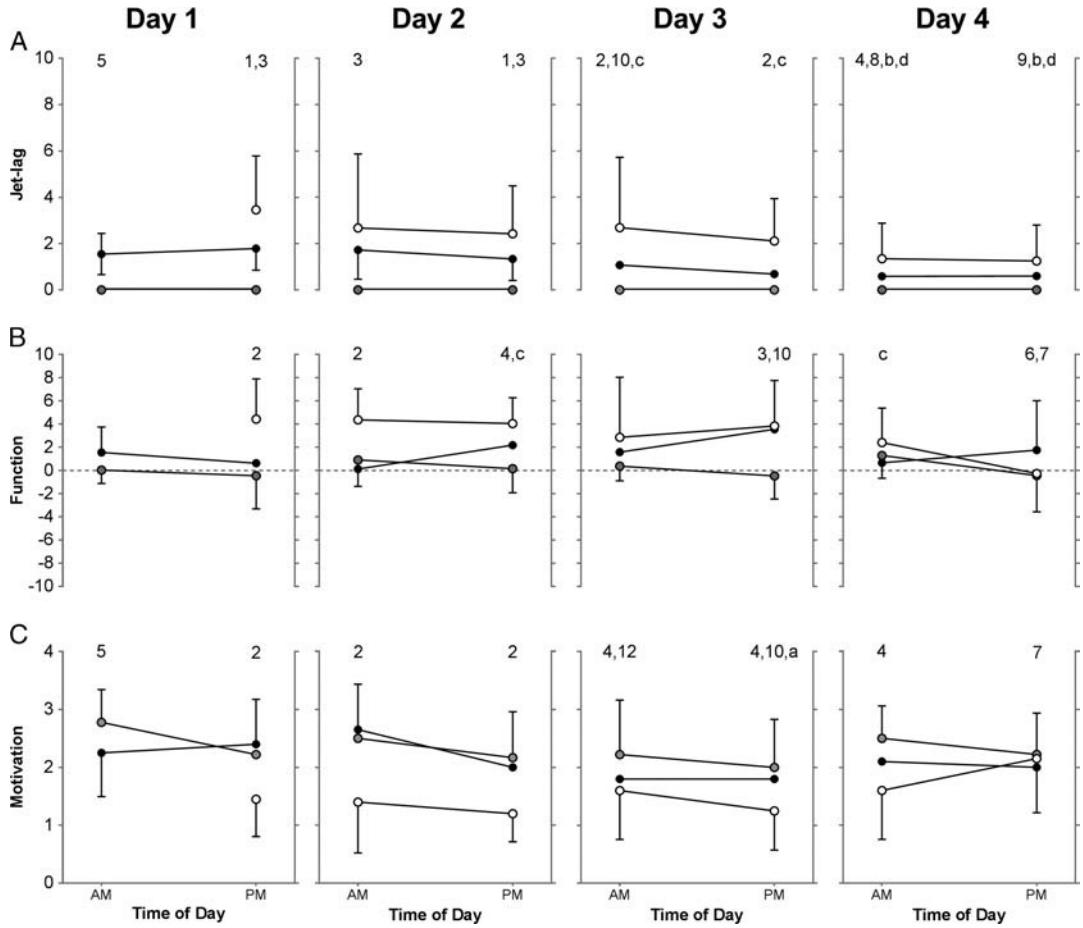

FIGURE 3—Mean ± SD for jet lag (A), function (B), and motivation (C) during BASE (gray circles), WEST (black circles), and EAST (white circles). [1]EAST significantly different from WEST ($P < 0.05$). [2]EAST significantly different from BASE and WEST ($P < 0.05$). [3]EAST and WEST significantly different from BASE ($P < 0.05$). [4]EAST significantly different from BASE ($P < 0.05$). [5]WEST significantly different from BASE ($P < 0.05$). [6]Significantly different from 0900 h in EAST ($P < 0.05$). [7]Significantly different from all other days in EAST ($P < 0.05$). [8]Significantly different from days 1 and 2 in EAST ($P < 0.05$). [9]Significantly different from day 1 in EAST ($P < 0.05$). [10]Significantly different from day 1 in WEST ($P < 0.05$). [11]Significantly different from day 2 in EAST ($P < 0.05$). [12]Significantly different from day 2 in WEST ($P < 0.05$). [a]Large ES indicating EAST is different from WEST ($d > 0.90$). [b]Large ES indicating EAST and WEST are different from BASE ($d > 0.90$). [c]Large ES indicating difference from all other days in BASE ($d > 0.90$). [d]Large ES indicating difference from days 1 and 2 in WEST ($d > 0.90$).

sleep ratings were worse on days 1 and 4 in BASE (3.7 ± 4.1 and 3.3 ± 4.8 AU) compared with WEST (0.7 ± 3.0 and 0.9 ± 3.2 AU, $d > 0.90$), on day 4 in BASE compared with EAST (0.8 ± 3.7 AU, $d = 1.11$), and on day 2 in EAST (4.8 ± 5.1 AU) compared with WEST (2.0 ± 2.5 AU, $d = 1.06$).

b. Day

Jet lag ratings were significantly lower on day 4 compared with days 1 and 2 in EAST ($P < 0.05$). Large ES also suggested that jet lag was lower on day 4 compared with days 1 and 2 in WEST ($P > 0.05$; $d > 0.90$). Sleep ratings were significantly better on day 4 compared with 2 ($P = 0.02$) in EAST.

PM testing (1700 h local time, ~0100 h WEST, ~0900 h EAST)

a. Travel direction

Jet lag was significantly worse in WEST compared with BASE on days 1 and 2 ($P < 0.05$) and EAST compared with BASE and WEST on days 1–3 ($P < 0.05$). Large ES indicated that jet lag was still elevated on day 4 in EAST and WEST compared with BASE ($d > 0.90$). Function ratings were significantly worse on day 1 in EAST compared with BASE and WEST, on day 2 in EAST compared with BASE, and on day 3 in EAST and WEST compared with BASE ($P < 0.05$). Large ES also indicated that function was worse on day 2 in WEST compared with BASE ($d = 1.32$). Diet ratings were significantly worse on days 1 and 2 in WEST (4.3 ± 3.2 and 4.0 ± 2.8) compared with EAST (2.2 ± 2.2 and 2.0 ± 1.7) and BASE (2.5 ± 1.5 and 2.2 ± 2.6; $P < 0.05$), and bowel movement ratings were significantly worse on day 1 in EAST compared with BASE (1.6 ± 2.5 vs 0.2 ± 0.3 AU, $P = 0.01$).

b. Day

Jet lag ratings were significantly lower on days 3 and 4 compared with day 1 in EAST ($P < 0.05$). Large ES also

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

suggested that jet lag was lower on day 4 compared with days 1 and 2 in WEST ($P > 0.05$; $d > 0.90$). Function ratings were significantly better on day 4 compared with all other days in EAST ($P < 0.01$), and on day 3 compared with day 1 in WEST ($P = 0.03$). Bowel movement ratings were significantly better on day 3 (0.2 ± 0.5 AU) and day 4 (0.2 ± 0.6 AU) compared with day 1 (1.6 ± 2.5 AU) in EAST ($P < 0.05$). Lastly, diet ratings were significantly better on days 3 and 4 (2.5 ± 2.2 and 2.2 ± 2.4 AU) compared with days 1 and 2 (4.3 ± 3.2 and 4.0 ± 2.8 AU) in WEST ($P < 0.01$).

c. Time of day

On day 4, function ratings were significantly better at 1700 h compared with 0900 h in EAST ($P < 0.01$).

Motivation, RPE, and physical feeling. Results for RPE and physical feeling are presented in Figure 2, with results for motivation presented in Figure 3. A significant main effect of travel direction was identified for motivation ($F_{2,190} = 34.12$, $P < 0.001$), with mean motivation across the 4 d significantly reduced in EAST compared with BASE and WEST, and WEST compared with BASE ($P < 0.05$). A significant main effect of travel direction was evident for physical feeling ($F_{2,70} = 3.97$, $P = 0.023$), with mean ratings across the 4 d significantly worse in EAST compared with BASE ($P = 0.007$).

AM testing (0900 h local time, ~1700 h WEST, ~0100 h EAST)

a. Travel direction

Motivation was significantly reduced on day 1 in WEST compared with BASE ($P = 0.04$), day 2 in EAST compared with BASE and WEST ($P < 0.05$), and days 3 and 4 in EAST compared with BASE ($P < 0.05$).

b. Day

Motivation was significantly lower on day 3 compared with day 2 in WEST ($P < 0.05$).

PM testing (1700 h local time, ~0100 h WEST, ~0900 h EAST)

a. Travel direction

Motivation was significantly reduced on days 1 and 2 in EAST compared with BASE and WEST ($P < 0.05$), and day 3 in EAST compared with BASE ($P < 0.05$). Large ES also suggested that motivation was reduced on day 3 in EAST compared with WEST ($d = 1.08$).

b. Day

Motivation was significantly lower on day 3 compared with day 1 in WEST ($P = 0.04$), and significantly better on day 4 compared with days 1–3 in EAST ($P < 0.05$). Physical feeling was significantly worse on days 1 and 2 in EAST compared with BASE ($P < 0.05$), and large ES indicated worse physical feeling in EAST on day 2 compared with WEST (d = 0.97) and on day 3 compared with BASE and WEST ($d > 0.90$). RPE was significantly higher on day 4 compared with day 1 at BASE ($P = 0.04$), with large ES suggesting that it was also higher on day 3 compared with day 1 ($d = 1.05$). Physical feeling was significantly better on day 4 compared with day 1 in EAST ($P = 0.01$), with large ES indicating that it was better on day 4 compared with all other days ($d > 0.90$).

c. Time of day

Motivation was significantly lower at 1700 h compared with 0900 h on day 2 in WEST ($P = 0.03$).

Hydration

Across the 4 d, mean USG was significantly lower in WEST (1.012 ± 0.006) compared with BASE (1.016 ± 0.007) and EAST (1.019 ± 0.007; $P < 0.05$), and BASE compared with EAST ($P < 0.05$). USG was significantly lower in WEST (1.014 ± 0.007) and EAST (1.014 ± 0.007) compared with BASE (1.019 ± 0.006) on day 1 ($P < 0.05$), BASE (1.015 ± 0.008 and 1.015 ± 0.007) and WEST (1.013 ± 0.006 and 1.011 ± 0.005) compared with EAST (1.020 ± 0.006 and 1.021 ± 0.007) on days 2 and 3 ($P < 0.05$), and WEST (1.018 ± 0.004) compared with BASE (1.017 ± 0.008) and EAST (1.018 ± 0.008) on day 4 ($P < 0.05$). USG was significantly greater on day 1 compared with days 2 and 3 in BASE ($P < 0.05$), but significantly reduced on day 1 compared with days 2 and 3 in EAST ($P < 0.05$).

DISCUSSION

The present study investigated the effects of 21-h air travel across eight time zones, both west and east, on sleep, subjective jet lag, fatigue and motivation, and physical performance related to team sports. Results indicated that west travel had negligible effects on sleep, fatigue, and intermittent sprint performance. Conversely, sleep quantity and intermittent sprint performance were reduced up to day 2, alongside increased perceived fatigue and reduced motivation up to day 4 after east travel. Irrespective of travel direction, reduced maximal sprint and CMJ performance were evident up to days 3 and 4 after travel, respectively. As such, the present study indicates performance deficits in lower-body power are evident up to 96 h after travel in either direction. Although within the initial 72 h, sleep, fatigue, and both maximal and intermittent sprint performance are disturbed to a greater extent after east travel. Accordingly, practitioners should be aware that 72 h after long-haul travel may be required before the restoration of optimal readiness to train or compete.

Training and competition can occur within 24–48 h after long-haul travel for elite team sport athletes. A novel finding of the present study was therefore the reduction of intermittent sprint performance on days 1 and 2 after east but not west travel. This is despite the time of day (1700 h local time) corresponding to more "unfavorable" body clock time in west compared with east travel (~0100 h vs ~0900 h on

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

day 1). Reduced and unchanged intermittent sprint performance has previously been reported on day 1 after 24 h of simulated travel, with sleep duration on the night of arrival the main difference between the two studies (13,14). Thus, reduced sleep duration observed in the first three nights after east travel could explain the reduction in YYIR1 performance in the present study. Comparable to previous simulated (13,14) and actual travel studies (15,37), subjective ratings of fatigue and motivation were exacerbated after travel in the present study, particularly east. In part, these altered subjective responses as a consequence of either circadian misalignment, sleep disruption, and/or the demands of travel could also explain the reduction of intermittent sprint performance (14,30). Although east and west travel had similar flight schedules and durations, the change in time zones meant a difference in local arrival time. This could also have contributed to the intermittent sprint performance reduction after east but not west travel, because there was a greater duration (~15 h) between arrival time and initial performance testing in west compared with east. Other detrimental consequences of travel that could affect performance are the low cabin humidity and potential risk of hypohydration (17). However, although USG across the 4 d was lower after west travel compared with baseline and east travel, results indicated that participants were adequately hydrated (USG <1.020) across all data collection periods (24). In summary, although west long-haul transmeridian travel with a morning arrival seems to induce negligible changes in intermittent sprint capacity, east travel with an evening arrival should be of concern to athletes and practitioners.

Reductions in CMJ performance were evident up to day 4 after both east and west travel in the present study. The timeline of recovery was different from the other measures of physical performance, with the greatest differences after both east and west travel compared with baseline on day 3, instead of day 1. Interestingly, the return of CMJ performance to baseline level coincided with greater motivation on day 4 at 1700 h compared with 0900 h local time after both east and west travel. In national team skeleton athletes, CMJ height was reduced after similar travel demands, but only on day 1 where testing occurred at ~0100 h body clock time (5). If circadian misalignment was the expected cause of CMJ performance impairments in the present study, (1) a change in the performance rhythm of a morning nadir and a late afternoon peak would have occurred, (2) east travel would have likely had a greater effect on performance compared with west travel, and (3) performance would have been reduced outside of its circadian peak window (22). Although testing after west travel at 0900 h (particularly day 1 after arrival) corresponded with the time of peak performance (~1700 h body clock time), there were negligible differences in CMJ performance between testing times (0900 h vs 1700 h) and travel directions (east vs west). The absence of these differences could be due to the nullification of circadian influence from a thorough warm-up (32,34), fatigue from travel, sleep disruption and/or previous exercise (34), or sensitivity of the CMJ test. Hence, although results from the present study suggest a 96-h recovery timeline for explosive lower-body power after both east and west long-haul travels, the time of day (local or body clock) does not seem to be influential.

Reduced 5- and 20-m sprint performance was also observed up to day 3 after both east and west travel. Similar to intermittent sprint performance, a greater reduction in 20-m sprint performance was detected on day 2 after east compared with west travel. This is again despite the time of day (1700 h local time) corresponding to a more unfavorable body clock time after west travel (~0100 h on day 1) and suggests that the time of day influence on performance posttravel may not be as integral as previously purported (25,27). In the same aforementioned group of skeleton athletes, no change in 30-m sprint performance was observed after similar travel demands (3), which was attributed to the competitive training environment and/or the athlete's high levels of motivation. Given the differences between testing and competition environments, motivational issues may explain the performance decrements noted in the present study (3,28,34). Regardless, compared with intermittent sprint capacity, the greater levels of neuromuscular fatigue, as indicated by reductions in both CMJ and 20-m sprint performance, are of concern for athletes after both east and west long-haul travel.

On the basis of an understanding of chronobiology, delayed sleep onset and early waking are anticipated after long-haul east and west travel, respectively (9,34,39). More prolonged sleep disruption after east travel is also expected, because circadian rhythms adjust quicker to a delay than an advance (9,34,39). To date, however, limited data on the effect of long-haul transmeridian travel on sleep patterns in passengers exist. Hence, a novel finding of the present study was that although there was no effect of west travel on posttravel sleep patterns, sleep duration was reduced for 3 d after east travel because of a later sleep onset and reduced time in bed. Conversely, in academics traveling to conferences, reduced sleep quantity and quality, and earlier sleep onset and waking were reported for 3 d after east travel, whereas delayed sleep onset and waking were reported for 5 d after west travel, with no changes in sleep quantity or quality (31). Considering that the delayed waking after west travel was attributed to the specific cohorts' lack of commitments (31), the anticipated early waking in west could have been masked in the present study by morning testing requirements, akin to training or competition demands. An average of only 6.5 h of sleep per night was obtained at baseline, which is recognized as below the recommended 7–9 h (19). Thus, a greater effect of travel, particularly west, may have been observed in individuals with better sleep habits. Subtle yet important differences in sleep architecture may also have been detected through polysomnography, which was unavailable in the present study (2,35). Lastly, because napping is a frequently reported symptom after long-haul transmeridian travel, it could also

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

be argued that different results may have been observed in the present study if the participants were not instructed to avoid napping.

A common occurrence after long-haul travel overnight is that on the night of arrival, the homeostatic factor of sleep regulation overrides the circadian factor and sleep quantity and/or quality increases (12,16,34). This is likely due to sleep disruption during travel, as a result of the flight schedule (e.g., enforced waking by a stopover) and/or an uncomfortable sleeping position (9,13,14). Compared with all subsequent days, increased sleep duration was evident on the night of arrival after west travel due to an earlier sleep onset and increased time in bed. Although it cannot be determined whether this was due to reduced sleep duration during travel, limitations exist for accurately assessing sleep during travel via actigraphy and/or self-report diaries. Conversely, sleep duration was reduced on the day of arrival after east travel due to an evening arrival (2220 h local time), which is typical due to an advance in time zones and has previously been reported to disrupt sleep in professional football players (11,15). Results from the present study suggest that a combination of travel demands, particularly the flight schedule and circadian misalignment, disrupted sleep after east travel.

The theoretical rate of realignment of circadian rhythms after transmeridian travel is half a day per hour of the time difference west and one day per hour of the time difference east (26). It was therefore anticipated that subjective jet lag ratings would be more prominent and persistent after east compared with west travel. Although this hypothesis can be accepted, the attenuation of jet lag ratings was faster than expected for both west (2 instead of 3–4 d) and east travel (4 instead of 7 d), which is consistent with data from other field-based studies (8,15). In the only other study to investigate the effect of travel direction on jet lag ratings, more prominent and persistent symptoms were also identified after east travel across eight time zones compared with west travel across six time zones in elite gymnasts (23). Furthermore, it was hypothesized that subjective jet lag would be worse at 0900 h after west travel and 1700 h after east travel, because this would have corresponded to ~0100 h body clock time on the first day after arrival. However, this was not consistently observed, which similar to CMJ performance could be because the circadian influence is nullified by other factors in field-based studies, such as fatigue from travel, sleep disruption, and/or previous exercise (34). Indeed, jet lag ratings are strongly associated with the level of perceived fatigue assessed at the same time (37). For example, increased subjective jet lag ratings were unexpectedly reported by professional football players for 5 d after long-haul travel north across one time zone (14). Results suggested that this was due to sleep disruption induced by an early departure time and evening competition, together with fatigue after competition rather than circadian rhythm misalignment (14).

Increased fatigue along with reduced motivation and tolerance to exercise was identified up to day 3 after east compared with west travel. These differences align with the reduced sleep duration observed for the first three nights after east travel and with previous observations that sleep disruption itself can exacerbate subjective ratings of fatigue, motivation, and tolerance to exercise (30). Together with the effect of sleep disruption on perceived fatigue, prolonged inactivity and exposure to mild hypoxia during air travel may also contribute (6). However, because the duration of travel was similar for east and west, any differences in perceived fatigue are likely to be attributed to circadian rhythm misalignment and/or sleep disruption. Previous research suggests that individuals are unable to maintain performance in sustained or repeated exercise bouts after sleep loss because of difficulties in maintaining the motivation to perform at a high intensity (27,30). Therefore, the cascade of travel-induced changes observed after east travel (reduced sleep duration together with exacerbated perceived fatigue and motivation) may explain the reduction of intermittent sprint performance.

In the present study, the participant population was relatively homogenous from a training and physical performance perspective, chronotype was accounted for, and a statistical technique that models for interindividual variation was used. Despite this, a common theme was the large interindividual variation in results, which may have masked potential differences between east and west travel. For example, when mean group reductions in performance were observed after travel, there were some individuals who performed better and *vice versa*. This variation in travel-induced responses could be due to age, flexibility of sleeping habits, and previous travel experience (15,38). In addition, the variation in motivation may have been larger in the present participant population compared with professional athletes (3,28,34). Regardless, significant interindividual variation in the disruption of body temperature circadian rhythms to 24 h of west travel across 10 time zones has previously been reported, with 50%–69% phase delays, 20%–38% phase advances, and 11%–19% unchanged (8). Thus, further research investigating the cause of this variation and using more progressive statistics to assess changes at an individual rather than a group level is warranted.

In conclusion, the present study is the first to compare the effect of east and west long-haul travel on integrated measures related to physical performance for team sports. Results suggest a greater effect of east travel on sleep, subjective jet lag, fatigue and motivation, and maximal and intermittent sprint performance, particularly within the first 48–72 h after arrival. Although despite including "pre-east" measures as an internal control, it is recognized as a limitation of the present study that it was not a counterbalanced design. It is also acknowledged that actual team sport performance requires more than activity from a single muscle or group of muscles, for example, neural control, central decision making, and motivation (28). Nonetheless, the present study provides coaches and practitioners with novel information on the timeline of recovery specific to team sport physical performance after east and west long-haul travel. These data

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

can help to inform decisions on travel schedules in relation to competition and when to return to full training after travel. For example, a practical recommendation from the present study would be that team sport athletes should aim to arrive a minimum of 96 h before competition after both east and west long-haul travel for optimal performance. However, because of their demanding schedules, this is not always feasible. Thus, future research developing interventions that have field-based efficacy for reducing the effect of long-haul transmeridian travel, particularly on sleep, is warranted.

The authors would like to thank the participants for their time and enthusiasm, and gratefully acknowledge the allocation of resources provided by the Aspire Zone Foundation, Aspetar Orthopaedic and Sports Medicine Hospital, and University of Technology Sydney in support of this project. The authors would also like to thank Adulaziz Farooq and Lauren Banting for their assistance with statistical analysis.

Declarations: No conflicts of interest are declared. The research was solely funded by the corresponding authors' institution. The results of this study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The present study results do not constitute endorsement by the American College of Sports Medicine.

REFERENCES

1. Atkinson G, Batterham AM, Dowdall N, Thompson A, van Drongelen A. From animal cage to aircraft cabin: an overview of evidence translation in jet lag research. *Eur J Appl Physiol*. 2014;114(12):2459–68.
2. Beaumont M, Batéjat D, Piérard C, et al. Caffeine or melatonin effects on sleep and sleepiness after rapid eastward transmeridian travel. *J Appl Physiol*. 2004;96(1):50–8.
3. Bullock N, Martin DT, Ross A, Rosemond D, Marino FE. Effect of long haul travel on maximal sprint performance and diurnal variations in elite skeleton athletes. *Br J Sports Med*. 2007;41(9):569–73.
4. Carnes AJ, Petersen JL, Barkley JE. Effect of peer influence on exercise behavior and enjoyment in recreational runners. *J Strength Cond Res*. 2016;30(2):497–503.
5. Chapman DW, Bullock N, Ross A, Rosemond D, Martin DT. Detrimental effects of west to east transmeridian flight on jump performance. *Eur J Appl Physiol*. 2012;112(5):1663–9.
6. Coste O, Van Beers P, Touitou Y. Impact of hypobaric hypoxia in pressurized cabins of simulated long-distance flights on the 24 h patterns of biological variables, fatigue, and clinical status. *Chronobiol Int*. 2007;24(6):1139–57.
7. Darwent D, Roach G, Dawson D. How well do truck drivers sleep in cabin sleeper berths? *Appl Ergon*. 2012;43(2):442–6.
8. Edwards BJ, Atkinson G, Waterhouse J, Reilly T, Godfrey R, Budgett R. Use of melatonin in recovery from jet-lag following an eastward flight across 10 time-zones. *Ergonomics*. 2000;43(10):1501–13.
9. Forbes-Robertson S, Dudley E, Vadgama P, Cook C, Drawer S, Kilduff L. Circadian disruption and remedial interventions: effects and interventions for jet lag for athletic peak performance. *Sports Med*. 2012;42(3):185–208.
10. Foster C, Daines E, Hector L, Snyder AC, Welsh R. Athletic performance in relation to training load. *Wis Med J*. 1996;95(6):370–4.
11. Fowler P, Duffield R, Howle K, Waterson A, Vaile J. Effects of northbound long-haul international air travel on sleep quantity and subjective jet lag and wellness in professional Australian soccer players. *Int J Sports Physiol Perform*. 2015;10(5):648–54.
12. Fowler PM, Duffield R, Lu D, Hickmans JA, Scott TJ. Effects of long-haul transmeridian travel on subjective jet-lag and self-reported sleep and upper respiratory symptoms in professional rugby league players. *Int J Sports Physiol Perform*. 2016;11(7):876–84.
13. Fowler PM, Duffield R, Morrow I, Roach G, Vaile J. Effects of sleep hygiene and artificial bright light interventions on recovery from simulated international air travel. *Eur J Appl Physiol*. 2015;115(3):541–53.
14. Fowler P, Duffield R, Vaile J. Effects of simulated domestic and international air travel on sleep, performance, and recovery for team sports. *Scand J Med Sci Sports*. 2015;25(3):441–51.
15. Fowler PM, McCall A, Jones M, Duffield R. Effects of long-haul transmeridian travel on player preparedness: case study of a national team at the 2014 FIFA World Cup. *J Sci Med Sport*. 2017;20(4):322–7.
16. Fullagar HH, Duffield R, Skorski S, et al. Sleep, travel, and recovery responses of national footballers during and after long-haul international air travel. *Int J Sports Physiol Perform*. 2016;11(1):86–95.
17. Hamada K, Doi T, Sakura M, et al. Effects of hydration on fluid balance and lower-extremity blood viscosity during long airplane flights. *JAMA*. 2002;287(7):844–5.
18. Hardy CJ, Rejeski WJ. Not what, but how one feels: the measurement of affect during exercise. *J Sport Exerc Psychol*. 1989;11(3):304–17.
19. Hirshkowitz M, Whiton K, Albert SM, et al. National Sleep Foundation's sleep time duration recommendations: methodology and results summary. *Sleep Health*. 2015;1(1):40–3.
20. Krustrup P, Mohr M, Amstrup T, et al. The Yo-Yo intermittent recovery test: physiological response, reliability, and validity. *Med Sci Sports Exerc*. 2003;35(4):697–705.
21. Lamond N, Petrilli RM, Dawson D, Roach GD. Do short international layovers allow sufficient opportunity for pilots to recover? *Chronobiol Int*. 2006;23(6):1285–94.
22. Leatherwood WE, Dragoo JL. Effect of airline travel on performance: a review of the literature. *Br J Sports Med*. 2012;47(9):561–7.
23. Lemmer B, Kern RI, Nold G, Lohrer H. Jet lag in athletes after eastward and westward time-zone transition. *Chronobiol Int*. 2002;19(4):743–64.
24. Popowski LA, Oppliger RA, Patrick Lambert G, Johnson RF, Kim Johnson A, Gisolf CV. Blood and urinary measures of hydration status during progressive acute dehydration. *Med Sci Sports Exerc*. 2001;33(5):747–53.
25. Reilly T, Atkinson G, Budgett R. Effect of low-dose temazepam on physiological variables and performance tests following a westerly flight across five time zones. *Int J Sports Med*. 2001;22(3):166–74.
26. Reilly T, Atkinson G, Edwards B, et al. Coping with jet-lag: a position statement for the European College of Sport Science. *Eur J Sport Sci*. 2007;7(1):1–7.
27. Reilly T, Piercy M. The effect of partial sleep deprivation on weight-lifting performance. *Ergonomics*. 1994;37(1):107–15.
28. Reilly T, Waterhouse J. Sports performance: is there evidence that the body clock plays a role? *Eur J Appl Physiol*. 2009;106(3):321–32.
29. Sargent C, Lastella M, Halson SL, Roach GD. The validity of activity monitors for measuring sleep in elite athletes. *J Sci Med Sport*. 2016;19(10):848–53.
30. Skein M, Duffield R, Edge J, Short MJ, Mündel T. Intermittent-sprint performance and muscle glycogen after 30 h of sleep deprivation. *Med Sci Sports Exerc*. 2011;43(7):1301–11.
31. Takahashi M, Nakata A, Arito H. Disturbed sleep–wake patterns during and after short-term international travel among academics attending conferences. *Int Arch Occup Environ Health*. 2002;75(6):435–40.
32. Taylor K-L, Cronin J, Gill ND, Chapman DW, Sheppard J. Sources of variability in iso-inertial jump assessments. *Int J Sports Physiol Perform*. 2010;5(4):546–58.

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

33. Thompson A, Batterham AM, Jones H, Gregson W, Scott D, Atkinson G. The practicality and effectiveness of supplementary bright light for reducing jet-lag in elite female athletes. *Int J Sports Med*. 2013;34(7):582–9.
34. Thun E, Bjorvatn B, Flo E, Harris A, Pallesen S. Sleep, circadian rhythms, and athletic performance. *Sleep Med Rev*. 2015;23:1–9.
35. Tonetti L, Pasquini F, Fabbri M, Belluzzi M, Natale V. Comparison of two different actigraphs with polysomnography in healthy young subjects. *Chronobiol Int*. 2008;25(1):145–53.
36. Vandenbogaerde TJ, Hopkins WG. Monitoring acute effects on athletic performance with mixed linear modeling. *Med Sci Sports Exerc*. 2010;42(7):1339–44.
37. Waterhouse J, Edwards B, Nevill A, et al. Do subjective symptoms predict our perception of jet-lag? *Ergonomics*. 2000;43(10):1514–27.
38. Waterhouse J, Edwards B, Nevill A, et al. Identifying some determinants of "jet lag" and its symptoms: a study of athletes and other travellers. *Br J Sports Med*. 2002;36(1):54–60.
39. Waterhouse J, Reilly T, Atkinson G, Edwards B. Jet lag: trends and coping strategies. *Lancet*. 2007;369(9567):1117–29.

Copyright © 2017 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.